

Prac 6

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB

# 1. Load the Iris dataset
df = pd.read_csv('Iris Dataset Prac (1,3.2,6,10).csv')

# 2. X = Features (all numeric columns), y = Target (Species)
X = df[['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']]
y = df['Species']

# 3. Split into 80% Training and 20% Testing
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42)

# 4. Initialize Gaussian Naive Bayes (used for continuous numeric data)
model = GaussianNB()

# 5. Train the model
model.fit(X_train, y_train)

# 6. Predict the classes for the test set
y_pred = model.predict(X_test)

from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score

# 1. Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)
```

2. Performance Metrics

Note: For multi-class (3 species), we use average='macro'

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accuracy = accuracy_score(y_test, y_pred)
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precision = precision_score(y_test, y_pred, average='macro')
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recall = recall_score(y_test, y_pred, average='macro')
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error_rate = 1 - accuracy
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print(f"\nAccuracy: {accuracy:.2f}")
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```
print(f"Error Rate: {error_rate:.2f}")
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```
print(f"Precision: {precision:.2f}")
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```
print(f"Recall: {recall:.2f}")
```